

Verification and Validation Matrix

Automated verification, human validation and evidence-governance matrix

ENTERPRISE REMEDIATION REPORT

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Evidence-led | contract-governed | accessibility-aware | rollout-ready

Truthful workflows. Explicit controls. Verifiable enterprise evidence.

Applies to: STQ-UIUX-REM-2026-08-06 **Rule:** verification proves the implemented system meets a specified contract; validation proves the resulting experience is usable, trustworthy, and operationally acceptable for its intended role. Neither substitutes for the other.

1. Evidence rules

- 1 Every run records commit SHA, environment, database/fixture version, browser/device, locale, currency, theme, role, package, timestamp, command, exit code, and artifact paths.
- 1 Tier-0 and Tier-1 evidence expires after 14 days or any relevant code/configuration change, whichever comes first. Tier-2 evidence expires after 30 days or a relevant change.
- 1 Screenshots without route, role, state, locale, currency, theme, and viewport metadata are illustrative, not certifying evidence.
- 1 A retry must preserve the failed evidence and state why the retry is legitimate. Retrying until green without root-cause classification invalidates the run.
- 1 A waived test is a failed gate until the named authority records scope, expiry, compensating control, rollback trigger, and follow-up owner.
- 1 Production telemetry can confirm safety and adoption; it cannot retroactively replace a missing pre-release control test.

2. Existing command baseline

<code>npm run typecheck</code>	Existing; passed 2026-08-06	Type-system integrity	Full log and exit 0	Every PR and phase gate
<code>npm run lint</code>	Existing; not rerun in this planning task	Current ESLint policy	Full log and exit 0	Every PR and phase gate
<code>npm test -- --runInBand</code>	Existing; full suite not rerun	Unit/integration regression	JUnit/log, failed-test details, coverage where applicable	Phase and release candidate
<code>npm run build:app</code>	Existing; not rerun	Production application build	Build log, bundle output, warnings disposition	Phase and release candidate
<code>npm run policy:gates</code>	Existing; not rerun	Current control-plane gates	Generated readiness artifacts and log	Phase and release candidate
<code>npm run verify:repo</code>	Existing; not rerun	Prisma, types, lint, policy, build, tests	Consolidated immutable run record	D-01
<code>npm run verify:release</code>	Existing; not rerun	Production-oriented release evidence	Consolidated immutable run record	D-05
<code>npm run module:surface:inventory</code>	Existing; current inventory revalidated	Module-surface current truth	Markdown/JSON inventory and count delta	G0-01, A-08
<code>npm run module:surface:fail</code>	Existing	No regression against approved module baseline	Inventory output and exit 0	A-08 onward
<code>npm run role:cockpit:gate</code>	Existing	Role cockpit readiness	Markdown/JSON output	B-02 onward
<code>npm run report:trust:export:gate</code>	Existing	Export/report truth policy	Markdown/JSON output	A-05 onward
<code>npm run workflow:assurance:runtime-check</code>	Existing	Workflow assurance runtime table	Log and exit 0	C/D gates
<code>npm run ui:smoke:public</code>	Existing; passed 6/6 on 2026-08-06	EN/FR public route reachability and screenshots	Six screenshot artifacts and JSON/log	A-06 onward; not accessibility certification
<code>npm run test:e2e</code>	Existing; not rerun	Current Playwright projects	HTML/JUnit report, trace/video on failure	D-01

The focused 2026-08-06 planning baseline also passed 6 Jest suites / 45 tests covering module entitlement, sidebar, dashboard route state, operating truth, UI route-smoke policy, and public landing content. This is baseline evidence only; it does not certify the proposed contracts.

3. Required gates to implement

Names below are reserved execution commands. They are **planned** until their work package adds them to `package.json` and publishes a schema/versioned artifact.

<code>npm run uiux:baseline:gate</code>	G0-01	Current-truth ledger	source scans, graph manifest, inventory	Counts classified; evidence fresh; no unexplained drift	Program Director	Correct inventory; never change thresholds to hide drift

<code>npm run uiux:fixtures:gate</code>	G0-03	Deterministic certification inputs	seeded tenants, roles, packages, locales, currencies, states	Repeatable hashes and tenant isolation	QE Lead	Rebuild fixture environment; preserve failed seed logs
<code>npm run inventory:summary:contract</code>	A-01	CT-01	inventory summary service/read model	Page-size invariant; reorder thresholds; as-of/scope/provenance; reconciliation exact	Inventory Lead	Disable new read path; show degraded state
<code>npm run read-state:contract</code>	A-02	CT-02	service results and page-state adapters	No unavailable/failed read renders as genuine empty/zero	Platform Architecture	Compatibility adapter retaining error semantics
<code>npm run display-context:gate</code>	A-03/A-04	CT-03	server context, UI, notifications, exports	EN/FR × XAF/XOF/EUR/USD correct; no unapproved literals/defaults	I18n Lead	Roll back surface flag; no USD fallback
<code>npm run action-truth:gate</code>	A-05	CT-06	control registry and action handlers	Every enabled control has executable outcome; export metadata complete	App Platform	Disable affected action with reason
<code>npm run accessibility:public-auth</code>	A-06/A-07	CT-07	public/auth route manifest	Zero critical/serious axe; keyboard complete; form errors associated; reflow pass	Accessibility Lead	Block rollout; revert layout only with semantic fixes intact
<code>npm run capability:parity:gate</code>	A-08	CT-04	route/action registry and package-role fixtures	UI visibility and server authorization agree for every protected fixture	Entitlements Lead	Return policy cohort to observe
<code>npm run route-registry:gate</code>	B-01	CT-05	Next route tree and typed registry	Every route classified; aliases redirect; no duplicate implementation	Web Platform	Restore alias redirect, not duplicate surface
<code>npm run route-state:gate</code>	B-03	CT-02/05/07	canonical route/state manifest	Required states declared, rendered, announced and correlated	Design Systems	Compatibility renderer preserving state truth
<code>npm run theme-contract:gate</code>	B-04/B-05	CT-08	tokens, CSS, screenshots	No new unapproved literals/islands; theme/contrast matrix passes	Design Systems	Token compatibility alias with expiry
<code>npm run onboarding:contract</code>	B-07	CT-09	signup/onboarding state machine	Minimal signup, resumability, idempotency, role-aware steps	Growth Product	Retain persisted progress across version fallback

npm run uiux:manifest:gate	B-06	CT-10	05_...MANIFEST.json, route registry, fixtures	Schema valid; every canonical protected route covered by risk policy	QE Lead	Block change/release until coverage is restored
npm run uiux:changed-route:gate	B-06	CT-10	git diff and manifest	Every changed Tier-0/1 route has required functional /a11y/visual evidence	QE Lead	No retry without cause; block merge
npm run uiux:copy:gate	C-00	CT-03/08	message catalog and UI source	No new unapproved literal copy; missing keys zero; pseudo-locale pass	Localization Lead	Revert copy migration or add governed key
npm run uiux:dialog:gate	C-00	CT-06/07	source and component tests	No native audited prompt/confirm; focus/reason/idempotency pass	Accessibility Lead	Use shared controlled dialog
npm run uiux:image:gate	C-00	CT-08	interactive image inventory	Dimensions/aspect/alt/offline policy complete; CLS within budget	Frontend Platform	Use known-dimension fallback
npm run uiux:module:certify -- --module <id>	C-01..C-06	CT-01..10 as applicable	module route subset and fixtures	Module DoD, route matrix, domain controls and screenshots pass	Domain Lead	Roll back module cohort/flag
npm run uiux:performance:gate	D-03	Performance SLO	production build and representative data	Approved route/bundle/interaction budgets, no material regression	SRE Lead	Roll back cohort; profile before retry
npm run uiux:release:gate	D-01/D-05	All	all evidence indexes and manifests	No open P0, all Tier-0/1 evidence fresh, signoffs and rollback ready	Independent Assurance	Release denied or cohort reversed

4. Work-package verification matrix

G0-01	Baseline counts, graph freshness, module inventory, finding delta	Audit author and architecture review discrepancies	Program Director	Gate 0
G0-02	ADR completeness and unresolved-decision gate	Product/control/security/design owner signatures	Principal Architect	Gate 0
G0-03	Fixture determinism, isolation, reset, state coverage	Domain owner verifies realistic scenarios without production data	QE Lead	Gate 0
A-01	CT-01 contracts, reconciliation, tenant/location isolation, failure injection	Inventory controller and accountant tie-out	Inventory Lead	A-P0
A-02	Completeness taxonomy contract and correlation propagation	Support recovery and control interpretation	Platform Architecture	A-P0
A-03/A-04	Context resolution, format matrix, hard-code ratchet	EN/FR linguistic and accounting display review	I18n Lead	A-P0
A-05	Action inventory, export artifact, retry/idempotency, audit trail	Download/evidence inspection	App Platform	A-P0
A-06/A-07	axe, keyboard, responsive/zoom, auth error semantics, funnel E2E	NVDA/VoiceOver, low vision, cognitive/mobile walkthrough	Accessibility Lead	A-Accessibility
A-08	Capability registry, role/package matrix, deep link/action parity	Security, commercial packaging, support review	Entitlements Lead	A-Entitlement
A-09	Redirect and action-disable assertions	Support link/action inspection	Web Platform	Phase A exit
B-01	Route inventory/registry parity and alias lifecycle	Support/analytics/documentation impact review	Web Platform	B-IA
B-02	Shortcut/nav capability snapshots and task ranking tests	Each primary persona reaches assigned work	Product Experience	B-Navigation
B-03	State registry and accessible renderer tests	Support/accessibility recovery walkthrough	Design Systems	B-State
B-04/B-05	Token ratchet, light/dark/system screenshots, contrast	Design + accessibility review at representative surfaces	Design Systems	B-Theme
B-06	Manifest schema, coverage, fixtures, trace/video capture	QE reviews reproducibility and flake	QE Lead	Phase B exit
B-07	Resume, idempotency, state migration, analytics	First-value usability study	Growth Product	B-Onboarding
C-00	Grid/dialog/image/copy component and source gates	Cross-domain primitive usability	Design Systems	C-Shared

C-01	Inventory module certification	Inventory/accounting workflows	Inventory Lead	C-Inventory
C-02	Purchasing/AP certification, three-way-match controls	Purchaser/AP workflow and control review	Purchasing/AP Lead	C-Purchasing
C-03	Finance/reconciliation/close certification	Accountant/external accountant close walkthrough	Accounting Controls	C-Finance
C-04	HR/payroll certification, privacy boundaries	Payroll operator/employee/manager journeys	Payroll Lead + Privacy	C-Payroll
C-05	Settings/admin certification and high-risk audit actions	Security/admin/support workflows	Administration Lead	C-Settings
C-06	POS desktop/tablet, offline/replay, cashier safety	Cashier/store-manager pilot	POS Lead	C-POS
C-07	Characterization parity, graph refresh, bundle boundaries	Maintainer/domain review	Principal Frontend Engineer	Phase C exit
D-01	verify:repo, policy, build, unit, E2E, manifest, changed-route gates	QE evidence audit	QE Lead	D-Automated
D-02	Validation protocols below	Independent role/domain/accessibility/localization signoffs	Validation Director	D-Human
D-03	Production build profiling, SLO, telemetry/alert tests	SRE operational review	SRE Lead	D-Operational
D-04	Cohort health gates and rollback drills	Support readiness and pilot acceptance	Release Director	D-Rollout
D-05	Evidence index integrity and open-risk check	Independent assurance decision	Assurance Lead	Final release

5. Human validation protocols

V-01 Accessibility	Accessibility specialist plus keyboard-only, NVDA/Windows and VoiceOver/macOS testers	Signup, login/error recovery, dashboard navigation, one CRUD form, one dense grid, one dialog, one export, one denial/partial state	Task completion without inaccessible workaround; WCAG 2.2 AA issues classified; zero open critical/serious	Script, recording/notes, issue links, signed result
V-02 Localization	Native/professional French reviewer plus domain SME	Public/auth, currency/date/number, state copy, finance/inventory/payroll terms, exports/notifications, pseudo-locale truncation	Meaning preserved, OHADA terminology approved, no clipping or ambiguous money	Glossary delta, screenshot set, signed checklist
V-03 Inventory controls	Inventory controller and accountant	Filtered KPI tie-out, pagination invariance, location scope, low-stock policy, failed/partial reads, export	Screen/API/export reconcile to source dataset and provenance	Reconciliation workbook/log and signoff
V-04 Purchasing/AP	Purchaser and AP clerk	Supplier-to-PO-to-receipt-to-invoice, approval denial, export	Status/control/action states truthful and recoverable	Scenario transcript and signoff
V-05 Finance/close	Accountant and external accountant	Payment, reconciliation exception, journal/close, stale/partial evidence, export	Evidence chain complete; role/privacy boundary correct	Close packet reference and signoff
V-06 Payroll/privacy	Payroll operator, employee, manager, privacy reviewer	Readiness, payroll run evidence, payslip self-service, manager scope, denial	Personal data minimized; scopes correct; outputs reconcile	Redacted evidence pack and signoff
V-07 POS	Cashier and store manager	Sale, void/return if supported, offline transition, replay/conflict, receipt	No duplicate/false completion; desktop/tablet task success within boundary	Pilot log, replay evidence, signoff
V-08 Administration	Owner/security admin/support	Module purchase/disable, user/permission change, audit reason dialog, deep link	Capability and server decision agree; support can explain/recover	Decision log and signoff
V-09 Onboarding	New owner and invited user cohorts	Minimal signup, interruption/resume, role-specific checklist, first task	No data loss; first actionable control visible; completion/adoption target met	Study notes and analytics report
V-10 Release operations	SRE, support, release manager	Feature flag change, alert, cohort pause, rollback, evidence lookup	Runbook executable within RTO; no data loss or policy bypass	Timed drill record and approvals

6. Failure classification and decision authority

Release blocker	Financial truth mismatch, cross-tenant leak, server/UI authorization disagreement, false completion, critical/serious accessibility issue on certified route	Stop rollout; contain; preserve evidence; execute rollback if exposed	Control/Security/Accessibility owner plus Release Director
Phase blocker	Contract, route, fixture, theme, or module gate failure	Stop affected workstream promotion; unrelated work may continue	Accountable work-package owner and independent reviewer
Advisory	Non-critical visual delta outside changed/risk-tier scope, non-regressing low-severity issue	Record with owner and expiry; may continue	QE Lead
Invalid test	Fixture/environment corruption proven by evidence	Preserve run, repair environment, rerun full affected scope	QE Lead, not feature implementer alone

7. Planning-task evidence and exclusions

Completed while producing this roadmap:

- 1 `npm run typecheck`: passed;
- 1 focused Jest: 6 suites / 45 tests passed;
- 1 `npm run ui:smoke:public`: 6/6 EN/FR routes passed with screenshots;
- 1 current-source inventories and audit deltas were revalidated.

Not executed because this task produced planning artifacts only: lint, build, full unit/integration suite, policy gates, authenticated E2E, three-browser matrix, screen-reader validation, production performance, rollout drill, and final release verification. They remain explicit blocking work in D-01 through D-05.